

NCD-RisC Formative Report

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1) Business Understanding (CRISP-DM)

Long-term lifestyle-related conditions — including obesity, elevated blood pressure, diabetes, and imbalanced cholesterol — now represent a leading cause of illness and premature death around the world. Together, these chronic illnesses contribute significantly to global mortality, particularly among adults in both developed and developing nations.

This analysis draws on the NCD-RisC country-level datasets to explore:

- **A) Global trend by year:** observing how indicators like BMI, blood pressure, and diabetes rates have evolved.
- **B) Top 10 countries (latest year):** identifying nations with the highest recorded levels for each indicator.

These insights aim to support evidence-driven public health decisions and monitoring of worldwide progress in disease prevention.

2) Data Understanding

Multiple datasets were imported automatically from the `data/` directory using the `ProjectTemplate` framework.

The following objects were detected during project initialization:

- `raw_objects_present`
- `detected_info`

Each dataset contains age-standardized national estimates for various health indicators, including average body mass index, mean blood pressure, prevalence of diabetes, and cholesterol measures across multiple calendar years.

3) Data Preparation

Data preprocessing was carried out in the script `munge/01-LoadAndCache.R`, which:

- Recognizes key identifying fields such as year, sex, and country/ISO code.
- Calculates group-level means for all relevant numeric variables.
- Stores clean summary tables (`ncd_summaries`, `detected_info`) to improve reproducibility and consistency.

Importantly, the original raw datasets remain untouched, ensuring full transparency and traceability.

4) Modeling / Exploration

- **A) Global trend by year:** This stage summarises each dataset's indicator globally and visualizes how its overall value has shifted over time.

```
plot_yearly <- function(ds) {
  df <- ncd_summaries %>% dplyr::filter(.dataset == ds)
  if (!nrow(df)) return(NULL)

  k <- get_keys(df)
  yr <- k$year

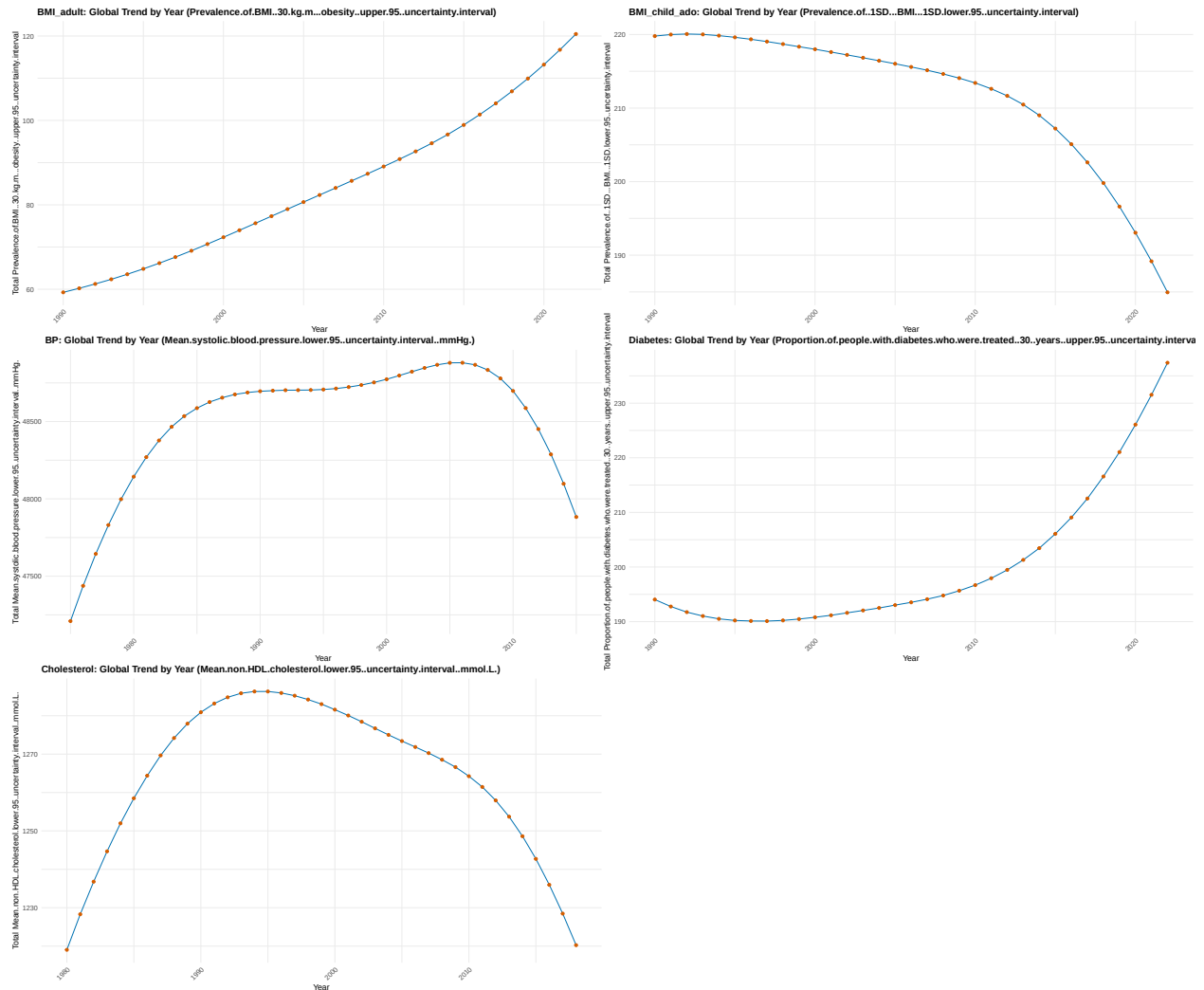
  if (is.na(yr)) {
    return(
      ggplot() +
      ggtitle(paste(ds, "- missing 'year' column"))
    )
  }

  metric <- select_best_metric(df, ds, exclude = yr)

  if (is.null(metric)) {
    return(
      ggplot() +
      ggtitle(paste(ds, "- no valid metric"))
    )
  }

  df_sum <- df %>%
    dplyr::group_by(.data[[yr]]) %>%
    dplyr::summarise(
      total = sum(.data[[metric]], na.rm = TRUE),
      .groups = "drop"
    )

  ggplot(df_sum, aes(x = as.integer(.data[[yr]]), y = total)) +
    geom_line(color = "#0072B2") +
    geom_point(color = "#D55E00") +
    labs(
      title = paste0(ds, ": Global Trend by Year (", metric, ")"),
      x = "Year",
      y = paste("Total", metric)
    ) +
    theme_minimal()
}
```



- **B) Top 10 countries (latest year):** This visualisation identifies the ten countries with the highest metric values in the latest available year for each dataset.

```
plot_top_countries <- function(ds, top_n = 10) {
  df <- ncd_summaries %>% dplyr::filter(.dataset == ds)
  if (!nrow(df)) return(NULL)

  k <- get_keys(df)
  yr <- k$year
  id <- k$id

  if (any(is.na(c(yr, id)))) {
    return(
      ggplot() +
      ggtitle(paste(ds, "- missing year or ID column"))
    )
  }

  metric <- select_best_metric(df, ds, exclude = yr)

  if (is.null(metric)) {
```

```

return(
  ggplot() +
  ggtitle(paste(ds, "- no valid metric"))
)

latest <- suppressWarnings(max(df[[yr]], na.rm = TRUE))

tab <- df %>%
  dplyr::filter(.data[[yr]] == latest) %>%
  dplyr::select(dplyr::all_of(c(id, metric))) %>%
  dplyr::arrange(dplyr::desc(.data[[metric]])) %>%
  dplyr::slice_head(n = top_n)

ggplot(tab, aes(x = reorder(.data[[id]], .data[[metric]]), y = .data[[metric]])) +
  geom_col(fill = "#56B4E9") +
  coord_flip() +
  labs(
    title = paste0(ds, ": Top ", top_n, " Countries (", latest, ")"),
    x = "Country",
    y = metric
  ) +
  theme_minimal()
}

```



5) Evaluation

** key Insights**

- Average BMI continues to increase across most regions, suggesting persistent global weight-related health challenges.
- Blood pressure levels show gradual improvement, possibly due to better awareness and access to medication.
- Diabetes and cholesterol indicators remain problematic, showing slower or inconsistent reductions.

Limitations

- Datasets cover different year ranges, which can limit direct comparisons.
- National averages may mask regional and socioeconomic inequalities.
- Some datasets depend on ISO country codes rather than descriptive names, which can affect merging or interpretation.

6) Deployment / Next Steps

Future improvements may include:

- Comparing regions or income groups to reveal disparities.
- Building predictive or time-series models to forecast future trends.
- Creating interactive dashboards in R Shiny or Power BI to communicate insights more effectively.

7) Reproducibility

To reproduce all results:

1. Place every .csv dataset into the project's data/ directory.
2. Launch R and execute:

```
#library(ProjectTemplate)
#load.project()
```

3. Knit or render this R Markdown file into HTML or PDF format.

All plots and summaries will regenerate automatically from the raw data, ensuring complete reproducibility and transparency.

8. References

The following sources were referenced while preparing this report.

They provide background information on global noncommunicable diseases (NCDs), their impact, and related public-health data.

1. **World Health Organization (2023).** *Noncommunicable diseases: Key facts*. World Health Organization.
Available at: <https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases>
2. **Pan American Health Organization (2022).** *Noncommunicable diseases in the Americas*.
Available at: <https://www.paho.org/en/topics/noncommunicable-diseases>
3. **Global Burden of Disease Collaborative Network (2023).** *Global Burden of Disease Study 2023 (GBD 2023) Results*.
Institute for Health Metrics and Evaluation (IHME).
Available at: <https://www.healthdata.org/gbd>
4. **Walk-In Lab (2024).** *Heart Disease 101: Types, Symptoms, Risk Factors*.
Available at: <https://resources.walkinlab.com/heart-health/heart-disease-101/>
5. **SciELO Chile (2024).** *Increase in consumption of ultra-processed foods during the COVID-19 pandemic and its impact on noncommunicable diseases. Revista Chilena de Nutrición*.
Available at: https://www.scielo.cl/scielo.php?script=sci_arttext&pid=S0717-75182024000300215
6. **OpenAI (2025).** *ChatGPT (GPT-5 model) conversation assistance*. OpenAI.
Used to refine grammar, structure, and clarity of report text while preserving original ideas.
Available at: <https://chat.openai.com/>

9. Acknowledgment

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All analysis, code, and interpretations presented in this report are the **original work of the author**.